Gambali Seshasai Chaitanya

Bachelor of Technology
Electronics and Communication Engineering
Indian Institute of Technology (BHU) Varanasi

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Education

Degree	Institute	Board / University	CGPA/Percentage	Year
B.Tech ECE	Indian Institute of Technology (BHU) Varanasi	IIT (BHU) Varanasi	8.43 (Till 7th Sem)	2022-2026
Senior Secondary	St. Francis School, Indirapuram	CBSE	96.8%	2021
Matriculation	St. Francis School, Indirapuram	CBSE	97%	2019

Experience

- **National University of Singapore** *January 2026 - Present*
  Visiting Student Researcher (with Prof. Flavien Solt) Singapore
  – Working towards developing FPGA-friendly approaches for microarchitectural side-channel analysis that eliminate the need for explicit Dynamic Information Flow Tracking.
- **University of California, Berkeley** *September 2025 - January 2026*
  Visiting Student Researcher (with Prof. Christopher Fletcher) California, USA
  – Engaged in hardware security research to mitigate microarchitectural side-channel attacks by adapting the *MileSan* dynamic analysis framework for FPGA-accelerated RTL simulations.
- **Jaguar Land Rover** *May 2025 - July 2025*
  Embedded Systems Intern Bengaluru, Karnataka
  – Designed and implemented a **secure two-stage bootloader architecture** with Root of Trust and Digital Signature verification, incorporating **AES decryption** capabilities to protect sensitive IP in compiled firmware binaries.
  – Optimized the bootloader design for low runtime memory usage and resource utilization for running in highly resource constrained environments.
- **RISC-V International @ Linux Foundation Mentorship 2025** *March 2025 - June 2025*
  Project Mentee Remote
  – Developed a **Hardware Abstraction Layer Transitional Library for Software to Hardware Posits For RISC-V** under the mentorship of Prof. Joshua Gyllinsky (also advised by Kurt Keville and Prof. John Gustafson), the RISC-V foundation. Selected as a mentee for the Spring term of the Linux Foundation Mentorship Program, Spring 2025
  – Improved and delivered a new version of the SoftPosit-cpp Library that provides native support for Posit arithmetic. Implemented math functions for 8, 16 and 32-bit posits with benchmarks, unit tests.
  – Created CI/CD pipelines for automating building and packaging, developed a Python Wrapper for SoftPosits and created a Debian Package.
- **Google Summer of Code, 2025** *June 2025 - September 2025*
  Software Development Intern Remote
  – Contributed to **OpenROAD**, an Open Source RTL2GDSII flow for digital IC design. Implemented end-to-end support for rectilinear floorplans inside the flow. Modified the user-facing tcl scripts, intermediate SWIG wrappers and the backend C++ logic to support polygonal floorplans.
- **Indian Institute of Technology Bombay** *May 2024 - July 2024*
  Summer Research Intern (with Prof. Sandip Mondal) Bombay, India
  – Developed Verilog and Embedded C-based applications using Vivado and Xilinx SDK to run on a Digilent Zybo FPGA board for developing an **FPGA-based controller** to perform Read, Write and Erase Operations on 2D and 3D SLC and MLC raw **NAND Flash** Chips.
  – Used bit-banging to precisely control the onboard GPIO pins to interface with the Raw NAND using the ONFI Consortium Protocol. Developed programs in Embedded C to run on **Embedded Linux** on an Altera DE1 SoC.
- **Google Summer of Code, 2024** *May 2024 - August 2024*
  Software Development Intern Remote
  – Contributed to **Apertium**, an open source machine translation platform. Developed core tools with command-line utilities for automated bidirectional dictionary generation from a set of parallel translated texts.
  – Experimented with and employed various methods for word alignment in parallel translated texts, such as Bayesian models with Markov Chain Monte Carlo and Graph Neural Networks for multiparallel word alignment.
- **University of Michigan** *November 2024 - March 2025*
  Software Development Intern (with Prof. Mehdi Saligane) Remote
  – Worked on creating an ABC script to improve synthesis options in Yosys for creating a balanced synthesis plan, balancing between Speed and Area optimized synthesis.

Journals

–Gambali Seshasai Chaitanya, Aditya D. Arkalgud, Shubham Pande, Ankit Arora, *"An Energy-Efficient, CMOS-Compatible Physical Reservoir Node with Post-Fabrication Tunable Decay Dynamics"*, IOP Neuromorphic Computing and Engineering (Accepted)

Conferences and Publications

–Gambali Seshasai Chaitanya, Aditya D. Arkalgud, Sourav De, Ankit Arora, *"Adaptive FeFET-Based LIF Neuron for Regression Tasks on Spiking Neural Networks"*, 7th International Conference on Emerging Electronics (ICEE 2025, Oral Presentation)

–João Pedro Cabral Miranda, Gambali Seshasai Chaitanya, Inbasekaran Perumal, Joshua V. Gyllinsky and Kurt Keville, *"Evaluating Numerical Formats for HPC Using Fractal Benchmarks"*, Conference for Next Generation Arithmetic (CoNGA'25, Poster Presentation)

–Gambali Seshasai Chaitanya, Ankit Arora, *"A Non-Volatile ReRAM and Subthreshold-FET-Based Kernel for Energy-Efficient Temporal Signal Processing Using Reservoir Computing"*, 23rd Non Volatile Memory Technology Symposium (NVMTS 2025, Poster Presentation)

–Gambali Seshasai Chaitanya, Arin Weling, Sandip Mondal, *"Stress Induced Failure Mechanism in NAND Flash Memory Devices,"* Electronic Materials Conference 2025 (Poster Presentation)

–Arin Weling, Gambali Seshasai Chaitanya, Sandip Mondal, *"Thermal-Induced Threshold Voltage Distribution Changes in 3D NAND Flash Cells: Experimental Characterization,"* Electronic Materials Conference 2025 (Oral Presentation)

–Asifa Khatun, Gambali Seshasai Chaitanya, Sandip Mondal, *"Page and Block-wise Program Disturb Operation in NAND Flash Memory,"* Electronic Materials Conference 2025 (Poster Presentation)

–Gambali Seshasai Chaitanya, Shubham Pande, Ankit Arora, *"ReRAM-based Crossbar Compatible Equality Checker for Neuromorphic Applications"*, 8th Students' Conference on Engineering & Systems (SCES-2024, Oral Presentation)

–Gambali Seshasai Chaitanya, Vaseekaran Elangovan, *"ReRAM Crossbar Compatible Efficient Equality Checker and Half Adder Implementation,"* XXII International Workshop on Physics of Semiconductor Devices (Oral Presentation)

Projects

- **RISC-V CPU Implemented in Verilog HDL** *Jan 2024 - Apr 2024*
  - Designed and implemented a 32-bit RISC-V CPU (RV32IM) in Verilog, validating architectural correctness via RTL testbenches and cross-compiled C-benchmarks using the riscv-gcc toolchain.
  - Currently executing the physical design flow for tape-out on the SkyWater 130nm node using the Cadence Digital Design Suite.
- **Neural Network Inference on FPGAs** *Jan 2024 - Apr 2024*
  Dr. Sonam Jain
  - Developed a Convolutional Neural Network and ran inference on FPGAs for Edge Computing
  - Implemented a behavioural model of a neuron in Verilog and used it to construct a fully connected network
- **Systolic Hardware Accelerator for Vector Matrix Operations** *March 2024*
  IChip, Udyam'24, IIT (BHU) Varanasi
  - Designed a systolic accelerator that efficiently performs Matrix multiplications and Convolutions
  - Orchestrated various components using an AXI-4 Interrupt controlled interface.

Achievements

- 3rd prize winner in the National Level Digital Design Hackathon conducted by the Ministry of Electronics & Information Technology of India (MeitY) under the Chips to Startup (C2S) Program.
- International Finalist in Google Code-In 2019, out of 14,798 participants from 117 countries
- Winner of the 'Startup with German' Scholarship, awarded by Goethe Institute. Felicitated by **Angela Merkel**
- Winner of Times Scholar 2019, Felicitated by the then Vice President of India, **Shri Venkiah Naidu**
- National Finalist in Hack AI, an AI/ML based Hackathon, Techfest, IIT Bombay
- 2nd in FPGA-based round in I-Chip, a Verilog-based event in Udyam'24, EES IIT (BHU) Varanasi
- Finalist in Digisim, a Digital Electronics Event, Udyam'24, EES IIT (BHU) Varanasi